

EDUCATION

University of Massachusetts Amherst 2021 – Present

M.S. & Ph.D. in Computer Science

Advisor: Donghyun Kim; Co-advisors: Sunghoon Ivan Lee, Joydeep Biswas

Hanyang University ERICA 2013 – 2019

B.S. in Robot Engineering (Cum Laude)

Exchange Student: The University of Texas at Austin, Dept. of ECE 2017 – 2018

HONORS AND AWARDS

- **Best Paper Finalist**, IEEE International Conference on Robotics and Automation (ICRA) 2026
- **Honourable Mention**, ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2026
- **Best Paper Finalist**, International Conference on Ubiquitous Robots (UR) 2024
- **Best Paper Award**, ACM SIGCHI Conference on Human Factors in Computing Systems (CHI) 2024
- **Robert and Deanna Hagerty Robotics Scholarship**, UMass Amherst 2024
- **Vision Assistance Race 2nd place**, CYBATHLON Challenges 2023
- **UMass Amherst CICS Jumpstart Fellowship**, UMass Amherst 2021

PUBLICATIONS

Conference

1. Synthetic Data Augmentation for Segmentation

Under review 2026

H. Hwang et al.

2. NavAble: A Large-Scale Dataset and Synthetic Data Generation Pipeline for Blind Navigation

The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS) 2026

H. Hwang[†], J.S. Monon[†], S. Yang, S.U. Patel, A. N.H. Nguyen, K. Nguyen, K. Garg, D. Gage, D. Hordofaa, A. Anjna, E. Ohn-Bar, D. Kim.

3. GuideTWSI: A Diverse Tactile Walking Surface Indicator Dataset from Synthetic and Real-World Images for Blind and Low-Vision Navigation

IEEE International Conference on Robotics and Automation (ICRA) 2026 [**Best Paper Finalist**]

H. Hwang, S. Yang, A. N.H. Nguyen, P. Goel, K. Adhikari, S.I. Lee, J. Biswas, N.A. Giudice, D. Kim

4. Human-Centered Development of Guide Dog Robots: Quiet and Stable Locomotion Control

IEEE International Conference on Robotics and Automation (ICRA) 2026 [**Oral**]

S. Yu[†], H. Hwang[†], T.M. Dang, J. Biswas, N.A. Giudice, S.I. Lee, D. Kim

5. GuideNav: User-Informed Development of a Vision-Only Robotic Navigation Assistant For Blind Travelers

ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2026 [**Honourable Mention, 3.6%**]

H. Hwang, S. Yang, J.S. Monon, N.A. Giudice, S.I. Lee, J. Biswas, D. Kim

6. Synthetic data augmentation for robotic mobility aids to support blind and low vision people

International Conference on Robot Intelligence Technology and Applications (RiTA) 2024

H. Hwang, K. Adhikari, S. Shodhaka, and D. Kim

7. **Is it safe to cross? Interpretable Risk Assessment with GPT-4V for Safety-Aware Street Crossing**
International Conference on Ubiquitous Robots (UR) 2024 **[Finalist, Top 8]**
H. Hwang, S. Kwon, Y. Kim, and D. Kim
8. **Towards Robotic Companions: Understanding Handler-Guide Dog Interactions for Informed Guide Dog Robot Design**
ACM Conference on Human Factors in Computing Systems (CHI) 2024 **[Best Paper Award, < 1%]**
H. Hwang, H.T. Jung, N.A. Giudice, J. Biswas, S.I. Lee*, and D. Kim*
9. **System Configuration and Navigation of a Guide Dog Robot: Toward Animal Guide Dog-Level Guiding Work**
IEEE International Conference on Robotics and Automation (ICRA) 2023
H. Hwang[†], T. Xia[†], I. Keita, K. Suzuki, J. Biswas, S.I. Lee*, and D. Kim*
10. **Control Scheme and Uncertainty Considerations for Dynamic Balancing of Passive-Ankled Biped and Full Humanoids**
IEEE-RAS International Conference on Humanoid Robots (Humanoids) 2018
D. Kim, S.J. Jorgensen, H. Hwang, and L. Sentis
11. **Computationally-Robust and Efficient Prioritized Whole-Body Controller with Contact Constraints**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2018
D. Kim, J. Lee, O. Campbell, H. Hwang, and L. Sentis

Journal

1. **ElderSim: A Synthetic Data Generation Platform for Human Action Recognition in Eldercare Applications**
IEEE Access 2021
H. Hwang, C. Jang, G. Park, J. Cho, and I.J. Kim
2. **Sensitive Capacitive Pressure Sensors over a Wide Pressure Range Enabled by the Hybrid Responses of a Highly Porous Nanocomposite**
Advanced Materials 2021
K.H. Ha, W. Zhang, H. Jang, S. Kang, L. Wang, P. Tan, H. Hwang, and N. Lu

RESEARCH EXPERIENCE

University of Massachusetts Amherst, Dynamic & Autonomous Robotics Lab 2021 – Present
Graduate Research Assistant

- Developing guide robots for blind and low-vision mobility through user studies, visual navigation, perception, and real-world deployment.
- Developing VLM/VLA methods and synthetic-data pipelines for safe and accessible navigation.

The University of Texas at Austin, Human Centered Robotics Lab 2017 – 2018
Undergraduate Research Assistant

- Testing and optimizing the 6 DOF passive-ankled biped robot, Mercury, for walking.

The University of Texas at Austin, Lu Research Group 2018
Undergraduate Research Assistant

- Manufacturing and testing flexible resistive force sensors for lower-limb prosthetic stress distribution.

WORK EXPERIENCE

Amazon – Frontier AI & Robotics, San Francisco, USA 2026 – Present
Member of Technical Staff Intern

- Developing map-conditioned VLA models for embodied nav and evaluating post-training strategies.

Glidance, Seattle, USA 2025 – 2026
PhD Research Intern

- Developed robust vision-based visual teach-and-repeat and VLM-based guidance for robot navigation.

Korea Institute of Science and Technology – Center for AI, Republic of Korea 2019 – 2020
Research Intern

- Real-time human action recognition system development and synthetic data augmentation evaluation.

INVITED TALKS

Human-Centered Embodied Intelligence for Safe and Accessible Real-World Autonomy
Texas Robotics, The University of Texas at Austin (Host: Joydeep Biswas) Oct. 2026

TEACHING EXPERIENCE

Teaching Assistant, UMass Amherst

- Reasoning Under Uncertainty (COMPSCI 240) – led discussion sessions Fall 2024
- Robotics (COMPSCI 603) – **mobile robot platform setup** Spring 2023
- Intro. to Robotics: Mechanics, Dynamics, & Control (COMPSCI 403) – **interactive quiz** Fall 2022

MENTORING EXPERIENCE

UMass Amherst (20+ M.S. and undergraduate students) 2023 – Present

- **Early Research Scholars Program:** Mentored 12 undergraduate researchers across three cohorts on perception and synthetic-data projects.
- **Honors Theses:** Mentored Krisha Adhikari (synthetic data) and Matthew Hersey (deep learning).
- **Research & Independent Study:** Mentored students in perception, navigation, audio tracking, and robot design; Shiven Patel won **1st Place at ACM TAPIA 2024**.

PRESS COVERAGE

- **Guide Dog Robot Development: Quiet and Stable Locomotion Control** May 2025
IEEE Spectrum Video Friday
- **Towards Robotic Companions: Understanding Handler-Guide Dog Interactions** May 2024
IEEE Spectrum Video Friday, UMass Amherst, Westside News (July 2022)

PROFESSIONAL SERVICE & OUTREACH

Student Representative, IEEE Robotics & Automation Society TC on Human-Robot Interaction 2026

Peer Review

- *Robotics* – ICRA'26, RA-L'25, UR'25, RiTA'24, Humanoids'24, RA-L'24, IROS'23, ICRA'22
- *Human-Robot Interaction* – HRI'26, TCDS'26, THRI'26, CHI'26, ACT'24, CHI'24

Outreach & Demonstrations 2022 – Present

Led autonomous guide robot demonstrations and robotics outreach for the Carroll Center for the Blind, Fidelco Guide Dog Foundation, Foundation Fighting Blindness, Polus Center, Massenberg STEM Institute, and Springfield High School of Science and Technology.

UMass Korean Graduate Student Association (KGSA) President 2022 – 2023